

Silas Kwabla Gah

Department of Computer Science, University of Ghana

Legon, Accra, Ghana

gahsilas@gmail.com | +233 558 075 023 | silsgah.github.io

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To: The Selection Committee

Visual Computing Section, Department of Applied Mathematics and Computer Science (DTU Compute)

Technical University of Denmark (DTU)

Richard Petersens Plads, Building 324

DK-2800 Kgs. Lyngby, Denmark

Re: Application for PhD Scholarship in Computer Vision and Multimodal Human-AI Interaction (Job ID: 7907)

Dear Members of the Selection Committee,

I am writing to submit my application for the 3-year PhD scholarship in Computer Vision and Multimodal Human-AI Interaction within the Visual Computing Section at DTU Compute. I hold an MPhil in Computer Science (120 ECTS equivalent) and am completing my doctoral research in multimodal vision-language models, 3D spatial grounding, and inference-time semantic adaptation. My academic background and research experience align directly with DTU's focus on developing AI systems capable of grounded, two-way visual communication between humans and machines.

Multimodal Vision-Language Models and 3D Visual Grounding

My recent research investigates methods for grounding multimodal large language models (MLLMs) in physical environments. In my latest study (*Gah et al., under submission to SACAIR 2026*), I designed a label-free process reward for Qwen2.5-VL using frozen 2D and 3D perception backbones (RegionPLC, SAM3, OpenScene). By introducing a two-arm validate-before-train protocol, we identified a spatial specification gaming failure mode and addressed it by formulating count-based precision and weighted $F_{0.5}$ objectives. This signal was then integrated into an online Group Relative Policy Optimization (GRPO) pipeline in VERL. This experience in combining generative language models with multi-view geometric representations directly supports the development of models that interpret visual cues and produce grounded visual feedback.

Probabilistic Calibration and Inference-Time Adaptation

In my doctoral research on Reliability-Calibrated Adaptive Semantic Arbitration (RCASA) (*under review at IEEE TPAMI*), I formulated a probabilistic framework to coordinate heterogeneous frozen foundation models at inference time ($\Delta\theta = 0$). Combining Platt confidence calibration with Bernoulli uncertainty estimation and contextual reliability weighting enables vision models to evaluate competing semantic hypotheses without parameter retraining. This approach achieved competitive results on open-vocabulary segmentation benchmarks (77.9% HM on PASCAL-5ⁱ and 33.68% HM on ScanNet200, under review at WACV 2026 and ACCV 2026). At DTU, I plan to apply these calibration and uncertainty mechanisms to ensure visual dialogue systems produce reliable, verifiable outputs.

Technical Background, University Teaching, and Research Experience

My qualifications and experience include:

- **Technical Implementation:** Proficient in Python, PyTorch, CUDA, vLLM, VERL, Hugging Face, SAM/SAM3, DINOv2, and multimodal LLM workflows.

- **University Teaching Experience:** Over 4 years as Assistant Lecturer at GIMPA, teaching Computer Vision, Deep Learning, and Data Structures, and supervising 55+ undergraduate and graduate research theses.
- **Research Output:** First-author manuscripts under review at peer-reviewed international venues (*IEEE TPAMI*, *SACAIR/RLVR*, *WACV*, *ACCV*), with a focus on publishing at top-tier vision conferences (CVPR, ICCV, ECCV).

I am prepared to relocate to Denmark, participate in teaching activities as a Teaching Assistant at DTU Compute, and complete the international research stay required by the PhD curriculum.

Thank you for considering my application. I look forward to the opportunity to discuss my research experience and proposed contributions with the selection committee.

Sincerely,

Silas Kwabla Gah